Andy Luo

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Education

Queen's University Kingston, ON
Bachelor of Computing (Hons.) | 4.2 GPA Expected May 2028

- **Specialization:** Computing and Mathematics
- **Relevant Coursework:** Data Structures and Algorithms, Software Specifications, Numerical Optimization for Artificial Intelligence, Linear Algebra, Multivariable Calculus

Skills

- **Languages:** Python, Java, JavaScript, JSX, C, C++, SQL, Bash
- **Frameworks and Libraries:** PyTorch, Scikit-Learn, React, Hydra-Core, OpenCV
- **Tools:** Weights & Biases, Docker, SLURM, ROS 2, Git

Experience

Undegraduate Student Researcher Kingston, ON
Royal Military College of Canada/Natural Resources Canada May 2026 – Present

- Awarded a highly competitive NSERC Undergraduate Student Research Award(USRA) valued at $10,000+ to conduct full-time laboratory research.
- Using PyTorch, developed a Temporal Convolutional Neural Network(TCN) architecture to accurately forecast geomagnetic conditions on Earth caused by solar winds.
- Engineered an end-to-end machine learning pipeline, including raw data preprocessing, robust feature engineering, model training, and rigorous evaluation.
- Developed an innovative loss function for time-series forecasting, addressing asymmetric loss and late-prediction penalties, currently being co-authored for publication.

Embedded Systems Intern Mississauga, ON
ApoSys Technologies Inc. April 2025 – August 2025

- Engineered a ROS2-based collision avoidance system for rail and road vehicles by fusing GPS, IMU, and Radar data.
- Developed an operational prototype through capable of obtaining highly accurate, real-time localization and environmental perception.
- Secured future funding by conducting a live demonstration of the functional system on a hi-rail truck for stakeholders at the Hudson's Bay Railway company.

Projects

ScreamStream | OpenCV, MediaPipe, Web Audio API, WebSockets

- Designed and implemented a real-time video control system that translates voice intensity and facial motion into video playback controls.
- Engineered a lightweight computer vision pipeline using OpenCV and MediaPipe for accurate head orientation estimation and facial gesture detection.

QHacks Hackathon 2026 | Python, Arduino, C++, React, Node.js

- Integrating an Arduino-based robotic mice capture vehicle with a custom React and Node.js web dashboard for real-time multi-sensor monitoring.

Interests

- Certified Hockey Referee, Hockey Coach and avid reader (Favourite novel: The Godfather)